

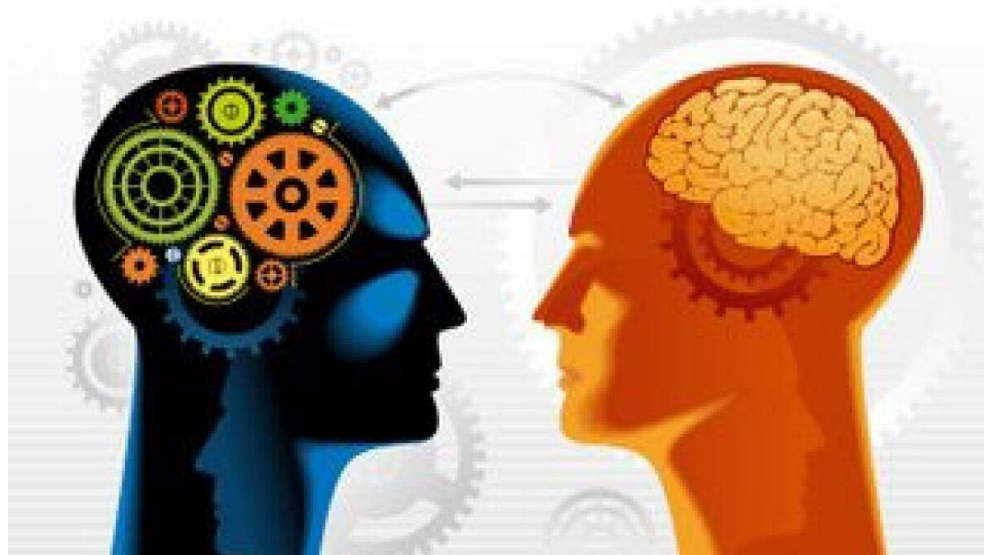
Linear Models and Extended Linear Models

Dr. Huseyin Kusetogullari

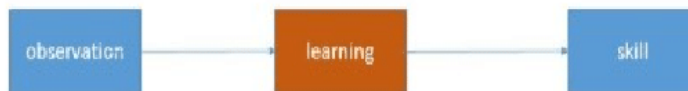
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- Linear Regression Models
 - The least square error
 - Univariate Linear Regression
 - Multi-variate Linear Regression
- Logistic Regression
- Support Vector Machines
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How Human learns?



• Human learning



• Machine learning



Types of Learning

- Unsupervised
- **Supervised**
- Semi-Supervised
- Reinforcement

Linear Models

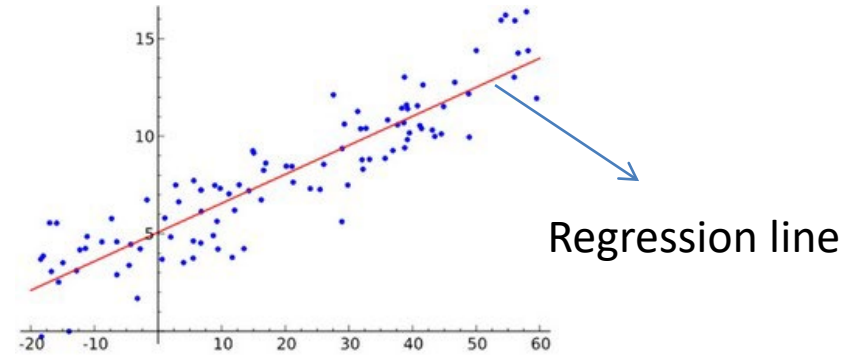
- Linear models are referred to the geometric models.
 - It could be described in 2-dimensional.
 - It could be described in 3-dimensional.
 - or in d-dimension.
- Geometric concepts such as lines and planes are used in this model, commonly called **linear models**.
- Supervised based methods.
- Linear models help to find the regression line which show the linear relationship between 2 or more than 2 variables.
- Linear model provides
 - Classification
 - Regression

Linear Models

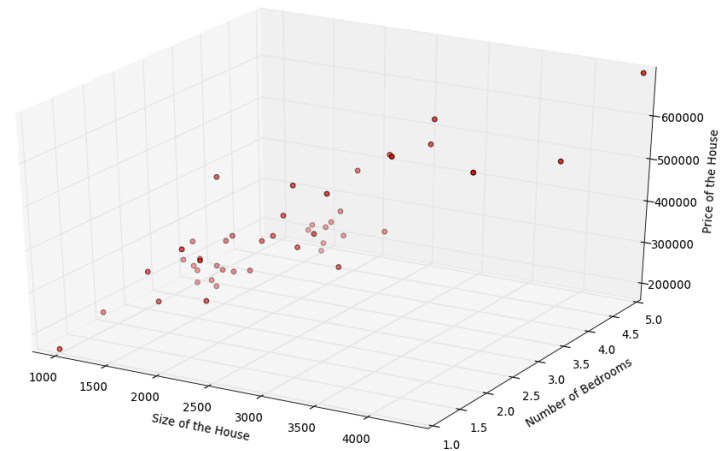
- There is a distinction between **classification** and **regression**, two particular instances of supervised learning.
 - **Classification** is when our labels yy can only take a finite set of values (categories or classes). Examples include:
 - Handwritten digit recognition: xx is an image with a handwritten digit, yy is a digit between 0 and 9.
 - Spam filtering: xx is an e-mail, and yy is 0 or 1 whether that e-mail is a spam or not.
 - **Regression** is when our labels yy can take any real (continuous) value. Examples include:
 - Predicting stock market.
 - Predicting sales.
 - Detecting the age of a person from a picture.

Linear Models

Example for 2-dimension
Linear

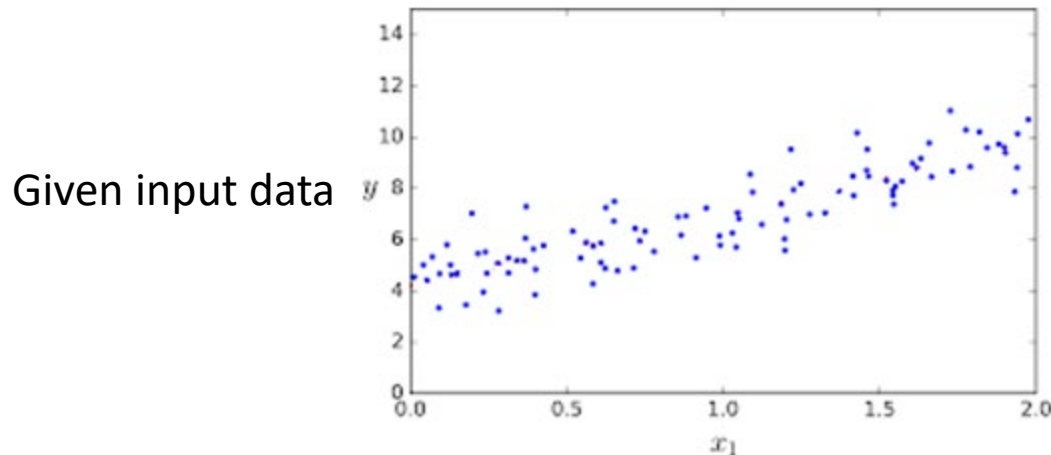


Example for 3-dimension
Plane



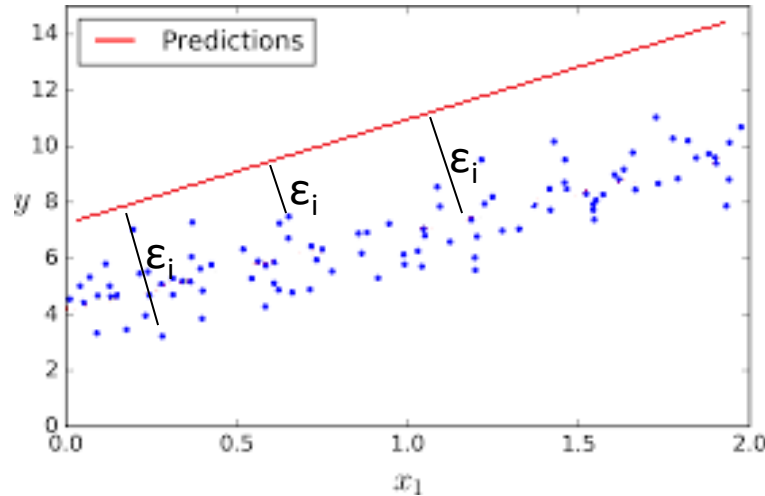
Linear models

- Simple comparing to the other machine learning methods.
- Consider that the input data is distributed linearly.
- The main aim is to find the best linear prediction model to predict the output data.



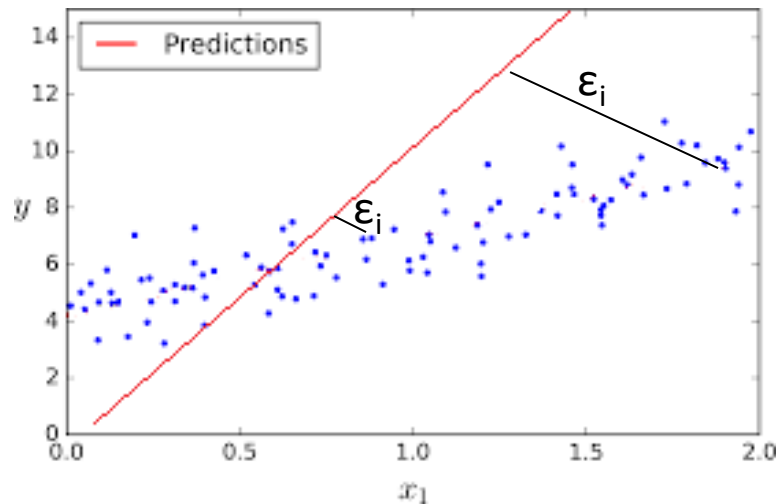
Linear models

Bad linear model



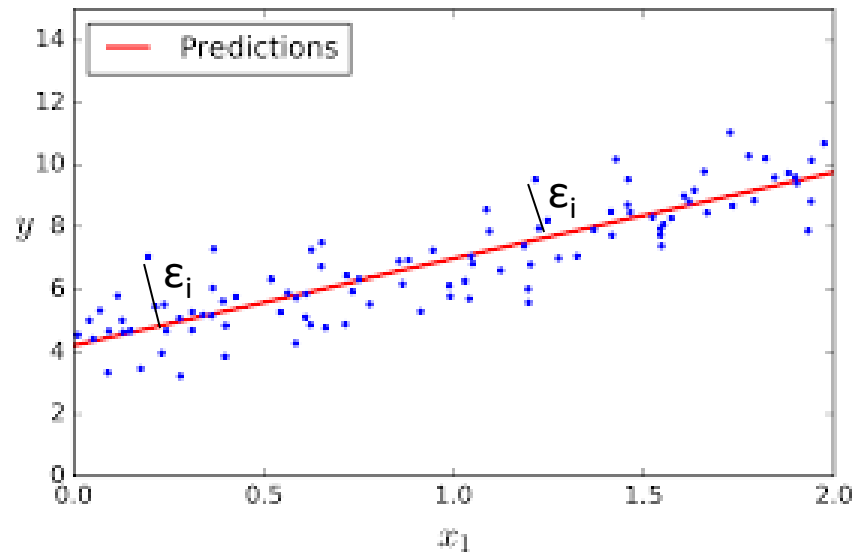
ϵ_i :error between
true and predicted value
of i^{th} sample

Bad linear model

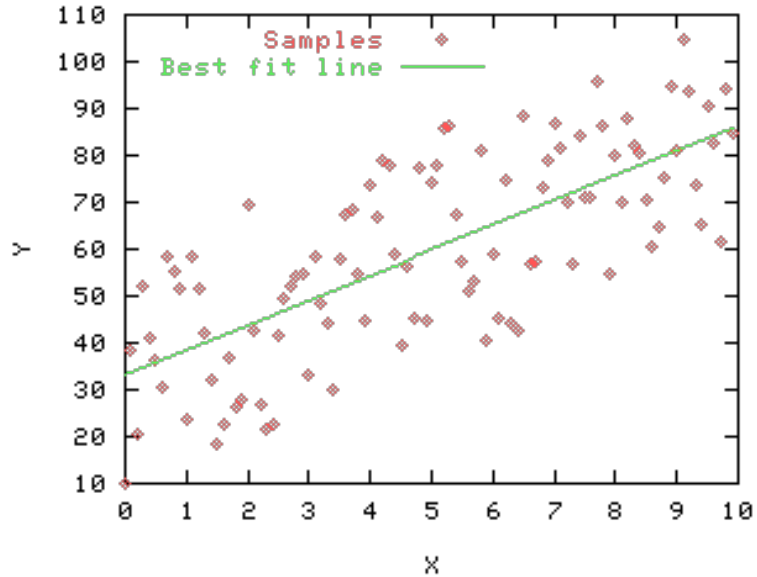


Linear models

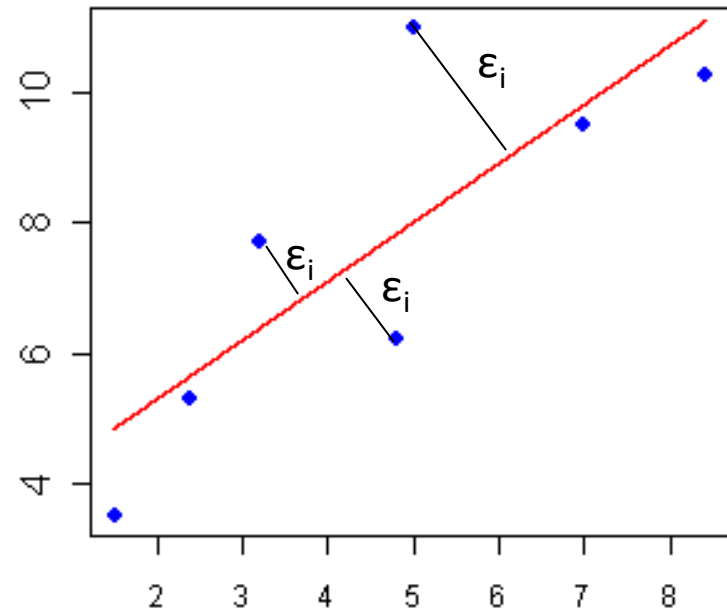
Best fit line



Linear models



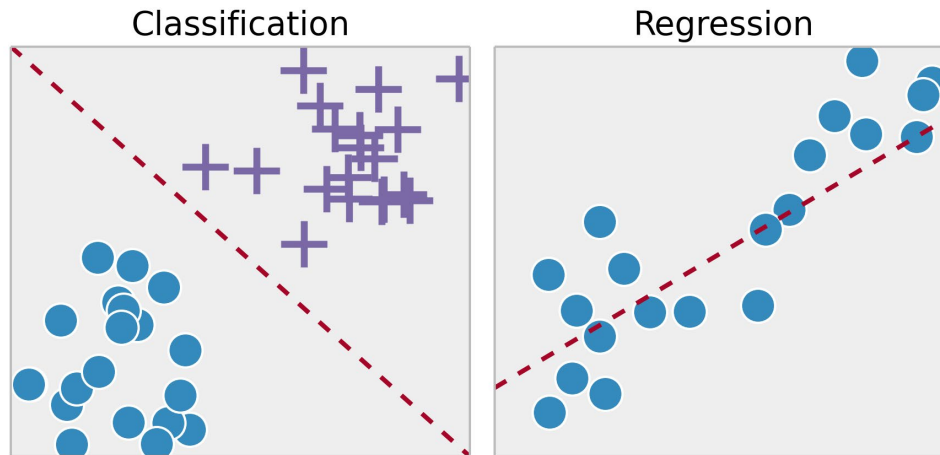
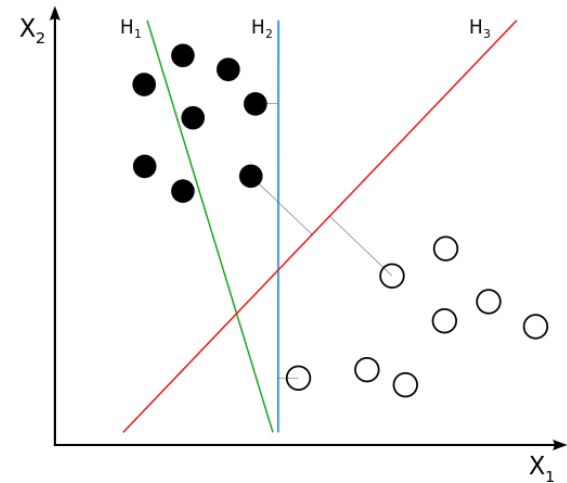
Linear Regression



Best fit line

Example

- It is necessary to provide good linear regression model.
- How can we obtain the regression model when we have d-dimensional data?



Linear models

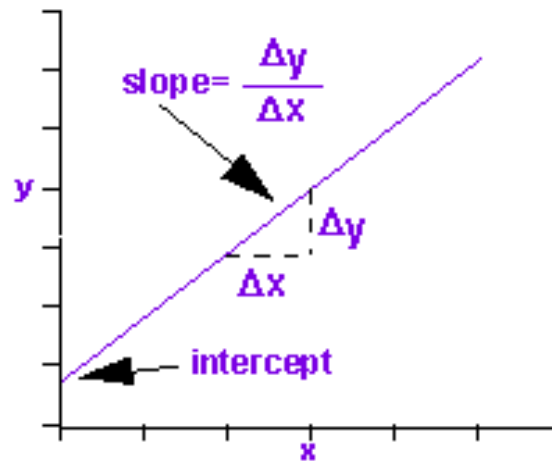
- y is called the dependent variable.
- x is called the independent variable.
- If y is a vector, we call this multiple regression.
- If x_1 and x_2 are two scalars or vectors of the same dimension and c_1 and c_2 are arbitrary scalars, then c_1x_1 and c_2x_2 is called a **linear combination** of x_1 and x_2 . If function f is a linear function of x , then

$$f(c_1x_1 + c_2x_2) = c_1f(x_1) + c_2f(x_2)$$

Linear models

- Linear functions take particular forms

$$f(x) = a + bx$$



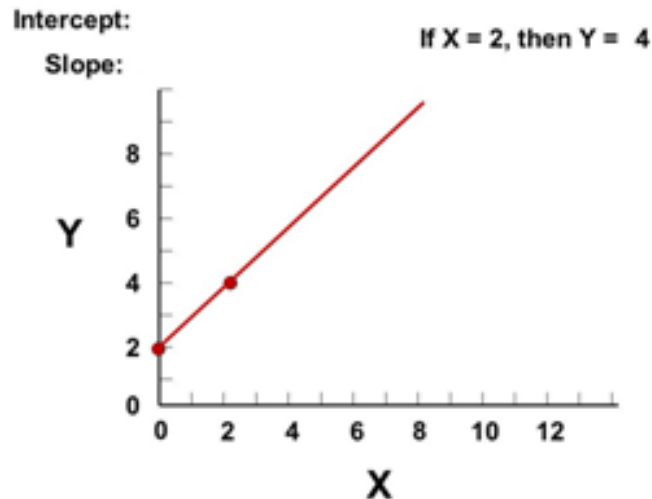
a: intercept

b: slope

function f depends on the variations of variable x

Example

Two data is given for (x, y) which are $(0, 2)$ and $(2, 4)$. Find the intercept and slope. Also find the linear model for this data.



Intercept: 2
Slope: 1

Linear models

If $\mathbf{x}=(x_1, x_2, x_3, \dots, x_d)$ is a vector and $f(\mathbf{x})$ is a scalar, then

$$f(\mathbf{x}) = a + b_1x_1 + b_2x_2 + b_dx_d = a + \mathbf{b}\mathbf{x}$$

with $\mathbf{b}=(b_1, b_2, b_3, \dots, b_d)$.

In most general, f is defined as:

$$f(\mathbf{x}) = \mathbf{M}\mathbf{x} + \mathbf{t}$$

where $\mathbf{M}$ is a d' -by- d matrix representing a **linear transformation**.

Linear models

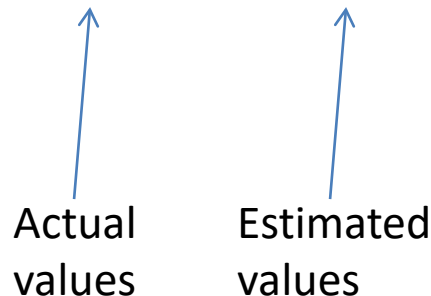
- **Parametric:** they have a fixed form with a small number of numeric parameters that need to be learned from data.
- **Stable:** Small variations in the training data have only limited impact on the learned model.
- **Underfitting:** The models are less likely to overfit the training data than some other models.
- Linear models can be used for all predictive tasks, including classification, probability estimation and regression.

The least squares method

- The method is used to learn linear models for classification and regression.
- The differences between the actual and estimated function values on the training examples are called **residuals** computed using:

$$\varepsilon_i = f(x_i) - f'(x_i)$$

Actual
values



Estimated
values

The least squares method

- The least squares method is first introduced by Carl Friedrich Gauss and it is defined as:

$$\sum_{i=1}^n \varepsilon_i^2 = \sum_{i=1}^n (f(x_i) - f'(x_i))^2$$

- The formulation computes the total error between the actual and estimated values and it is necessary to minimize this to find the best linear model.

Univariate Linear Regression

- Univariate refers to an expression, equation, function or polynomial of only one variable.
- Lets predict the shoe size of a person of a certain height. Our intuition tells us that there is a connection between these two *variables linearly*. Therefore, if we know the height of the person, we can predict the shoe size of that person.
- In case of relationship between blood pressure and age: the bigger value of one variable the greater value of another one, where the association could be described as *linear*.
- Both of these examples can very well be represented by a *linear regression model*, considering the mentioned characteristic of the relationships.

Univariate Linear Regression

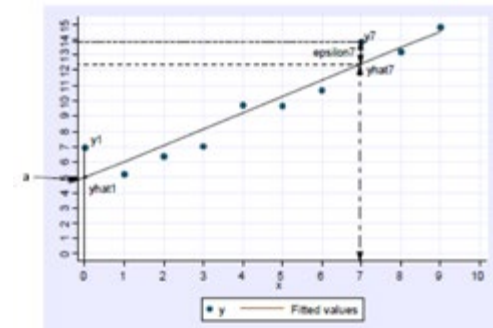
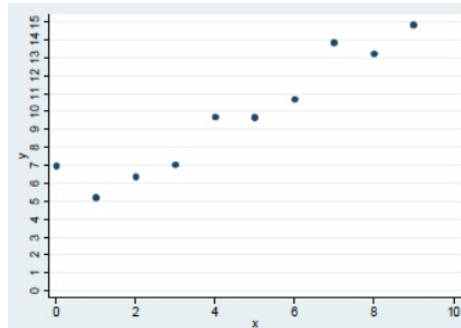
- Linear relationship between x and y (dependent variable)
- Epsilon describes the amount of error between x and y values.
- Minimize the sum of all squared deviations from the line

$$y_i = a + bx_i + \varepsilon_i$$

- This is done mathematically by the program

$$y_i = a + bx_i + \varepsilon_i \longrightarrow \varepsilon_i = y_i - a - bx_i$$

- The values of the dependent variable (values on the line) are called predicted values of the regression ($\hat{y}$): 4.97, 6.03, 7.10, 8.16, 9.22, 10.28, 11.34, 12.41, 13.47, 14.53 – these are very close to the true values; the estimated $a = 4.97$ and $b = 1.06$



Univariate Linear Regression

- Ordinary least squares regression: minimizes the squared residuals

$$y_i = a + bx_i + \varepsilon_i \quad \longrightarrow \quad \varepsilon_i = y_i - a - bx_i$$

$$\sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n \varepsilon_i^2$$

- The main aim is to minimize the ε_i^2

Univariate Linear Regression

- The relationship between x and y is described by the function:

$$y_i = a + bx_i + \varepsilon_i \quad \longrightarrow \quad \varepsilon_i = y_i - a - bx_i = S(a, b)$$

- Minimizing the function requires to calculate the first order conditions with respect to “ a ” and “ b ” and set them zero:

Univariate Linear Regression

- Take derivative of equation with respect to a

$$I: \sum_{i=1}^n (y_i - a - b * x_i) = 0 \Rightarrow -n a + \sum_{i=1}^n (y_i - b * x_i) = 0$$

$$\Rightarrow n a = \sum_{i=1}^n (y_i - b * x_i) \Rightarrow a = \bar{y} - b * \bar{x}$$

Average of y values

Average of x values

Univariate Linear Regression

- and b

$$\Pi: \sum_{i=1}^n (y_i - a - b * x_i) x_i = 0$$

$$\Rightarrow \sum_{i=1}^n y_i x_i - a x_i - b x_i^2 = 0 \quad a = \bar{y} - b * \bar{x}$$

$$\Rightarrow \sum_{i=1}^n y_i x_i - (\bar{y} - b \bar{x}) x_i - b x_i^2 = 0$$

$$\Rightarrow \sum_{i=1}^n y_i x_i - \bar{y} x_i + b \bar{x} x_i - b x_i^2 = 0 \Rightarrow \sum_{i=1}^n (y_i - \bar{y} + b \bar{x} - b x_i) x_i = 0$$

$$\Rightarrow \sum_{i=1}^n y_i - \bar{y} + b(\bar{x} - x_i) = 0 \Rightarrow \sum_{i=1}^n (y_i - \bar{y}) = -b \sum_{i=1}^n (\bar{x} - x_i)$$

$$\Rightarrow b = \frac{\sum_{i=1}^n (y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})}$$

$$\Rightarrow b = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = X'X^{-1}X'y$$

Univariate Linear Regression

- The regression line summarizes the linear relationship between two different variables. Regression coefficient or slope b' is found as follows

$$b = \frac{\sigma_{xy}}{\sigma_{xx}} = \frac{\text{cov}(x, y)}{\text{var}(x)} = \frac{\frac{1}{n-1} \sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1} \sum_i^n (x_i - \bar{x})^2} \quad \begin{matrix} \sigma_{xx} = \sigma_x^2 \\ \text{(Variance of variable x)} \end{matrix}$$

- Univariate linear regression assumes that a linear equation is $\bar{y} = \bar{a} + b'\bar{x}$

Using the above equation

$$\bar{a} = \bar{y} - b'\bar{x}$$

Intersect average of y slope average of x

Example

Lets assume that following input data (n=10) is given:

Age(x_i)	weight(y_i)
1	7
2	9
3	9
4	8
4	9
5	11
5	10
6	13
6	14
7	13

a) Find the Regression line with respect to age.

$$b = \frac{\sigma_{xy}}{\sigma_{xx}} = \frac{\text{cov}(x, y)}{\text{var}(x)} = \frac{\frac{1}{n-1} \sum_i^n (x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1} \sum_i^n (x_i - \bar{x})^2}$$

$$\bar{a} = \bar{y} - b' \bar{x}$$

Example

$$\sum x_i = 43 \rightarrow \bar{x} = \frac{43}{10} = 4.3$$

$$\sigma_{xx} = \frac{1}{N-1} \sum (x_i - \bar{x})^2 = \frac{32.1}{9} = \underline{\underline{3.55}}$$

$$\sum y_i = 102 \quad \bar{y} = \frac{102}{10} = 10.2$$

$$\sigma_{xy} = \frac{1}{N-1} \sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{37.4}{9} = \underline{\underline{4.155}}$$

$$\sum x_i^2 = 217 \quad \sum y_i^2 = 1094$$

$$b' = n \frac{\sigma_{xy}}{n\sigma_{xx}} = \frac{\sigma_{xy}}{\sigma_{xx}} \quad b' = \frac{4.155}{3.55} = 1.165$$

$$a = \bar{y} - b'\bar{x} \longrightarrow a = 10.2 - 1.165 \cdot 4.3 = \mathbf{5.19}$$

The regression line is $y = 5.19 + 1.165x$

Example

- Predict the value of y if x is 3.

$$y' = 5.19 + 1.165x = 5.19 + 1.165 \cdot 3 = 8.685$$

$$\rightarrow \varepsilon_i = f(x_i) - f'(x_i) = 9 - 8.685 = 0.315$$

Residual error

- Predict the value of y if x is 8.

$$y' = 5.19 + 1.165x = 5.19 + 1.165 \cdot 8 = 14.51$$

- Predict the value of y if x is 60.

$$y' = 5.19 + 1.165x = 5.19 + 1.165 \cdot 60 = 75.09$$

Age(x_i)	weight(y_i)
1	7
2	9
3	9
4	8
4	9
5	11
5	10
6	13
6	14
7	13

Multivariate Linear Regression

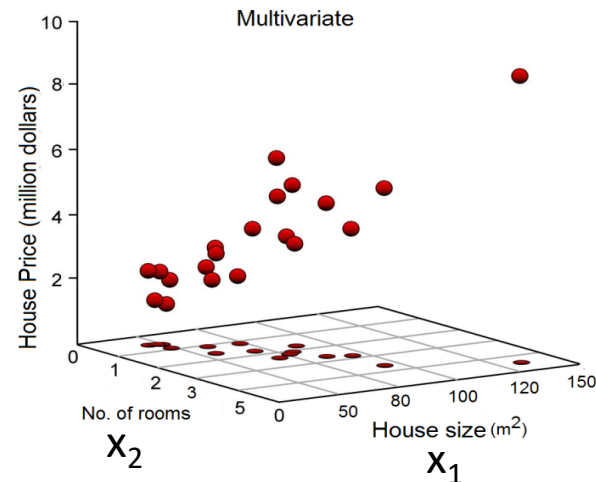
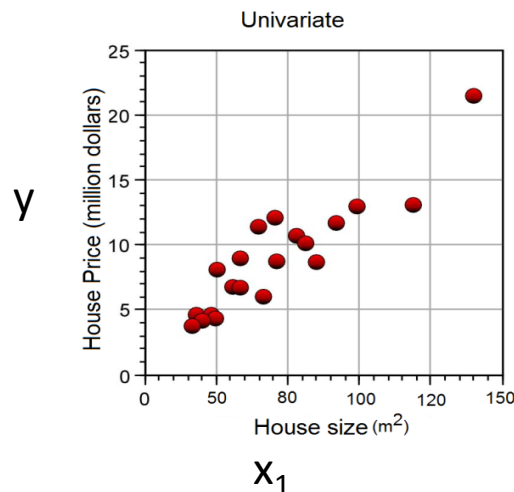
- Multivariate Linear Regression model is influenced by more than one variable with a linear relationship.
- Thus, a regression model is called the *multivariate linear regression*.

Multivariate Linear Regression

- This is similar to the linear regression model, but with **multiple independent variables**.
- Hence multivariate linear regression is formed as:

$$Y_i = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

- Where y is the dependent variable with n independent variables.



Multiple Regression Model in Matrix Form

- In matrix notation the multiple regression model is: $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

where

$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}, \quad \boldsymbol{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1k} \\ 1 & X_{21} & & X_{2k} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & & X_{nk} \end{bmatrix}$$

- Note, $\mathbf{Y}$ and $\boldsymbol{\varepsilon}$ are $n \times 1$ vectors, $\boldsymbol{\beta}$ is a $(k+1) \times 1$ vector and $\mathbf{X}$ is a $n \times (k+1)$ matrix.

Multivariate Linear Regression

- For multivariate case, we need to find the covariance between every feature and the target variable.

$X^T y \longrightarrow$ n-vector holding all the required covariances

$$x \longrightarrow (y_{1j}, \dots, y_{nj})$$

$$y \longrightarrow (y_{1j}, \dots, y_{nj})$$

Multivariate Linear Regression

- Let suppose that success of a student depend on three independent variables such as
 - IQ,
 - Giving time (hour) for study in a day
 - reading (number of words in minute).

Y	X_1	X_2	X_3
Student grade	IQ	Time for study	Speed of reading
56	122	8	120
46	118	5	123
91	134	14	151
39	102	5	92
61	98	13	119

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

Cont.

- Multivariate regression problem is

$$\hat{w} = S^{-1} X^T y$$



Scaling matrix

- The equation can be written as

$$\hat{w} = (X^T X)^{-1} X^T y$$

- $(X^T X)^{-1}$ acts as a transformation that decorrelates, centres and normalizes the features.

$$Y = X\hat{w} + \epsilon$$

Example1:

- $x_1 = [1, 3, 4, 7, 9, 9]$
- $x_2 = [9, 9, 6, 3, 1, 2]$
- $y = [3, 5, 6, 8, 7, 10]$
- Find a multivariate regression line. Find the errors with regarding to x_1 and x_2 .

Answer for Example1

$$x_1 = [1, 3, 4, 7, 9, 9]$$

$$x_2 = [9, 9, 6, 3, 1, 2]$$

$$y = [3, 5, 6, 8, 7, 10]$$

$$x_1 = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 7 \\ 9 \\ 9 \end{bmatrix} \quad x_2 = \begin{bmatrix} 9 \\ 9 \\ 6 \\ 3 \\ 1 \\ 2 \end{bmatrix} \quad y = \begin{bmatrix} 3 \\ 5 \\ 6 \\ 8 \\ 7 \\ 10 \end{bmatrix}$$

Answer for Example1

$$X = \begin{bmatrix} 1 & 1 & 9 \\ 1 & 3 & 9 \\ 1 & 4 & 6 \\ 1 & 7 & 3 \\ 1 & 9 & 1 \\ 1 & 9 & 2 \end{bmatrix} \quad Y = \begin{bmatrix} 3 \\ 5 \\ 6 \\ 8 \\ 7 \\ 10 \end{bmatrix}$$

Matlab Code for calculation:

```
X=[1 1 9;1 3 9; 1 4 6;1 7 3;1 9 1;1 9 2]  
Y=[3;5;6;8;7;10]  
(X'*X)^(-1)*X'*Y
```

Python Code for calculation:

```
import numpy as np  
X=np.matrix([[1, 1, 9], [1, 3, 9], [1, 4, 6], [1, 7, 3], [1,9,1], [1,9,2]])  
Y=np.matrix([[3],[5],[6],[8],[7],[10]])  
w=np.linalg.inv(X.T*X)*X.T*Y
```

$$\hat{w} = (X^T X)^{-1} X^T y$$

$$w = \begin{bmatrix} -1.6927 \\ 1.0990 \\ 0.4297 \end{bmatrix}$$

$$Y = X\hat{w} + \epsilon$$

$$Y = -1.6927 + 1.0990X_1 + 0.4297X_2$$

Answer for Example1

$$Y = -1.6927 + 1.0990 \begin{bmatrix} 1 \\ 3 \\ 4 \\ 7 \\ 9 \\ 9 \end{bmatrix} + 0.4297 \begin{bmatrix} 9 \\ 9 \\ 6 \\ 3 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3.2736 \\ 5.4716 \\ 5.2815 \\ 7.2894 \\ 8.6280 \\ 9.0577 \end{bmatrix}$$

$$y = \begin{bmatrix} 3 \\ 5 \\ 6 \\ 8 \\ 7 \\ 10 \end{bmatrix}$$

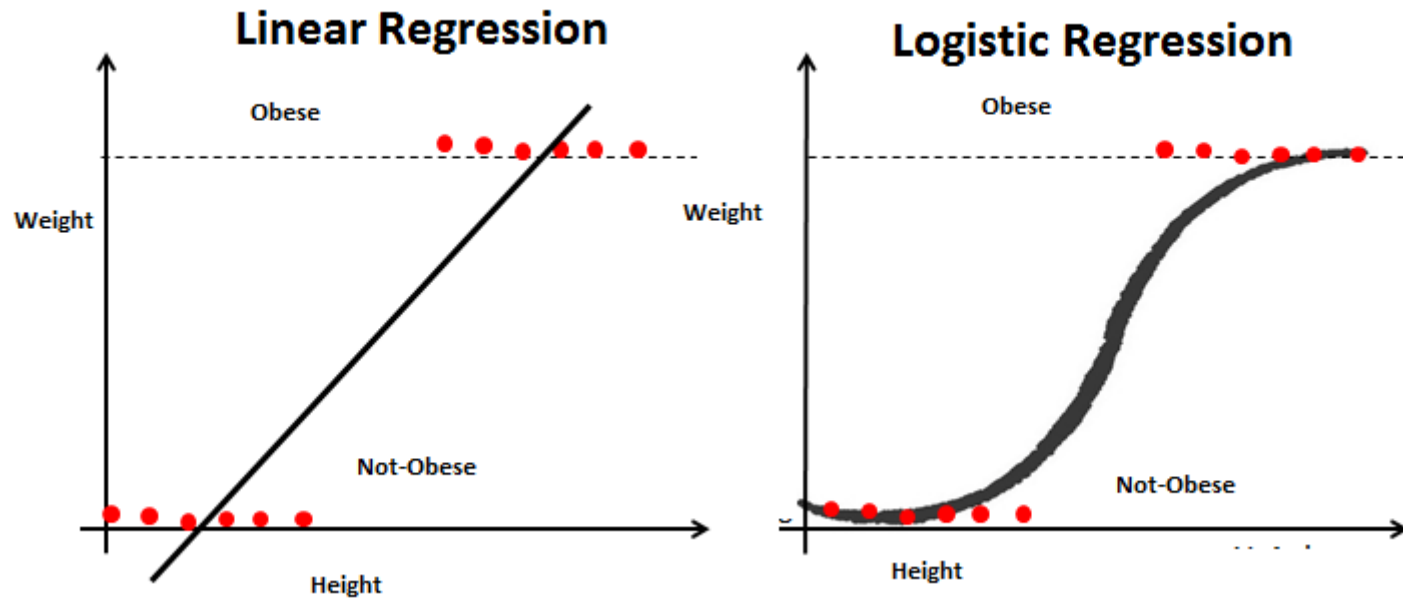
Exercise:

- Given is the following data:

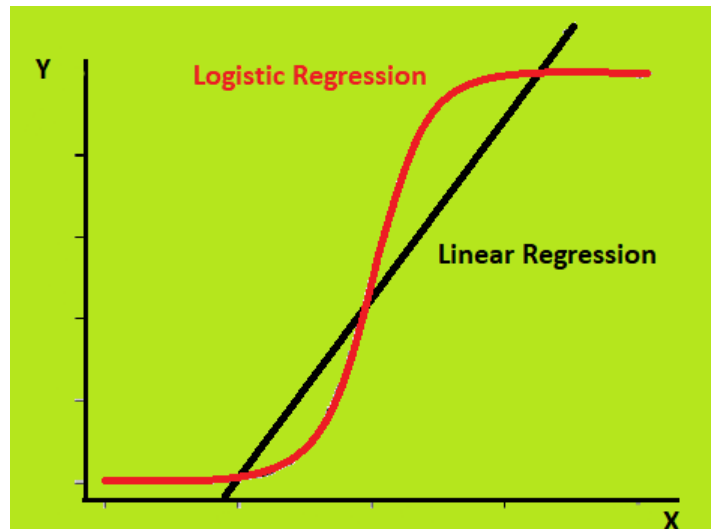
x_1	x_2	y
1	2	3
2	3	2
4	1	7
5	5	1

- Find a y value for $x_1 = 3$, $x_2 = 4$.

Linear Regression vs Logistic Regression



Linear Regression vs Logistic Regression



Logistic Regression

- Logistic regression is used to model the probability of a certain class or event existing such as pass/fail, win/lose, alive/dead or healthy/sick.
- For example,
 - To predict whether an email is spam (1) or (0)
 - Whether the tumor is malignant (1) or not (0)

Logistic Regression

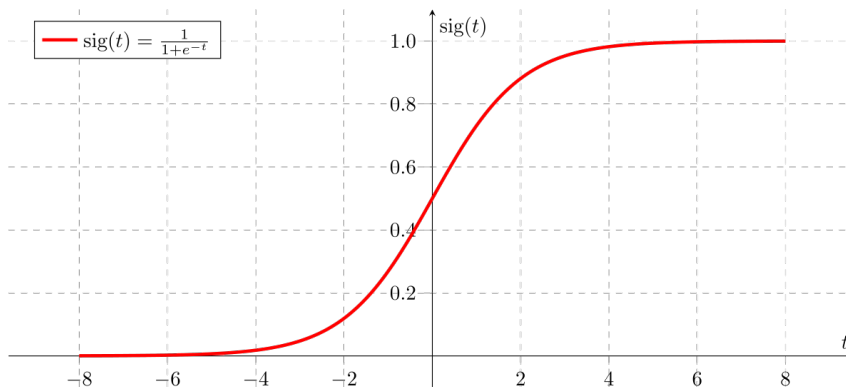
Model

If 'Z' goes to infinity, Y(predicted) will become 1 and if 'Z' goes to negative infinity, Y(predicted) will become 0.

Output = 0 or 1

Hypothesis $\Rightarrow Z = WX + B$

$h_{\Theta}(x) = \text{sigmoid}(Z)$



Logistic Regression

- Logistic Function

$$y = \frac{1}{1 + e^{-x}}$$

Linear Function

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2$$

- Logistic Regression equation

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2)}}$$

where p is the predicted output, B_0 is the intercept, and B_1 and B_2 are the coefficients for X .

Analysis of the hypothesis

- The output from the hypothesis is the estimated probability.
- This is used to infer how confident can predicted value be actual value when given an input X .
- Example,
 - Based on the x_1 value, let's say we obtained the estimated probability to be 0.8. This tells that there is 80% chance that an email will be spam.

Types of Logistic Regression

- 1. Binary Logistic Regression
 - The categorical response has only two possible outcomes. Example: Spam or Not
- 2. Multinomial Logistic Regression
 - Three or more categories without ordering. Example: Predicting which food is preferred more (Veg, Non-Veg, Vegan)
- 3. Ordinal Logistic Regression
 - Three or more categories with ordering. Example: Movie rating from 1 to 5

Important

- *In logistic regression, the outcome is always a categorical variable.

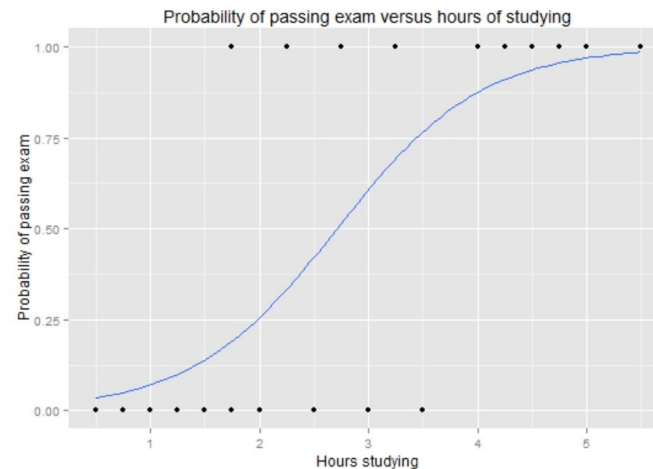
Example

- A group of 20 students spends between 0 and 6 hours studying for an exam. How does the number of hours spent studying affect the probability of the student passing the exam?

Hours	0.50	0.75	1.00	1.25	1.50	1.75	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	4.00	4.25	4.50	4.75	5.00	5.50
Pass	0	0	0	0	0	0	1	0	1	0	1	0	1	0	1	1	1	1	1	1

- Answer:

	Coefficient
Intercept	-4.0777
Hours	1.5046



Example

Intercept = -4.0777 and Hours = 1.5046 .

$$\text{Probability of passing exam} = \frac{1}{1 + \exp(-(1.5046 \cdot \text{Hours} - 4.0777))}$$

For example, for a student who studies 2 hours, entering the value in the equation gives the estimated probability of passing the exam of 0.26:

$$\text{Probability of passing exam} = \frac{1}{1 + \exp(-(1.5046 \cdot 2 - 4.0777))} = 0.26$$

For example, for a student who studies 4 hours, entering the value in the equation gives the estimated probability of passing the exam of 0.87:

$$\text{Probability of passing exam} = \frac{1}{1 + \exp(-(1.5046 \cdot 4 - 4.0777))} = 0.87$$

Support Vector Machines

- In machine learning, Support Vector Machine (SVM) is a non-probabilistic, linear, binary classifier used for classifying data by learning a hyperplane separating the data.
- Supervised based method.
- Linearly separable data admits infinitely many decision boundaries that separate the classes.
- The main aim of the algorithm is to keep the sum of the margin between classes and decision boundary line as large as possible.
- The training examples nearest to the decision boundary are called **support vectors**.
- The decision boundary of a support vector machine is defined as a linear combination of the support vectors.

Applications

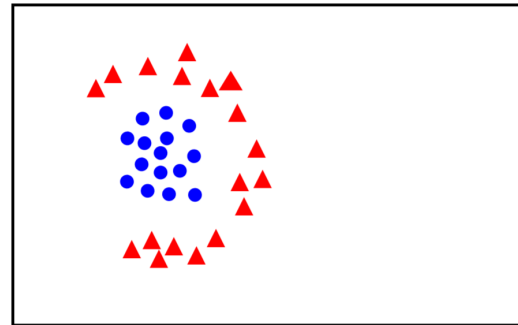
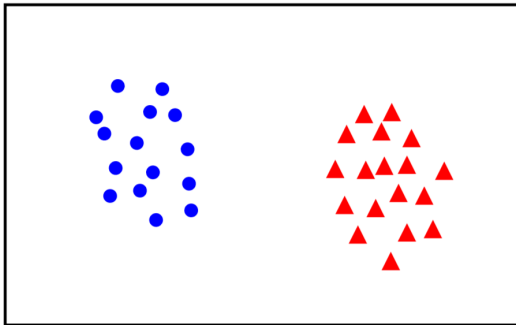
- SVMs can be used to solve various real-world problems:
 - SVMs are helpful in text and hypertext categorization
 - Classification of objects on the images.
 - Hand-written character and digit recognition using SVM.
 - Furthermore, the SVM algorithm has been widely applied in the biological, finance and other sciences.

Binary Classification

Given training data $(\mathbf{x}_i, y_i)$ for $i = 1 \dots N$, with $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \{-1, 1\}$, learn a classifier $f(\mathbf{x})$ such that

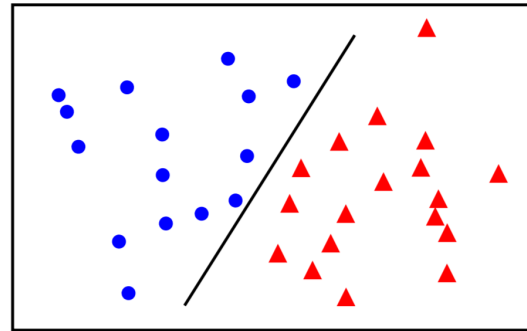
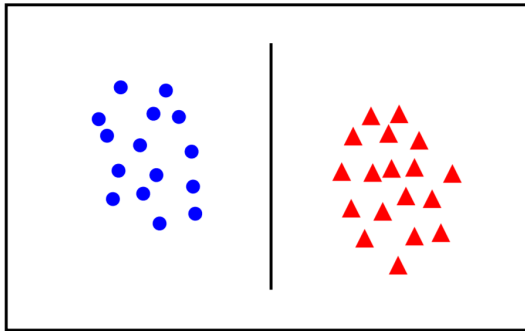
$$f(\mathbf{x}_i) \begin{cases} \geq 0 & y_i = +1 \\ < 0 & y_i = -1 \end{cases}$$

i.e. $y_i f(\mathbf{x}_i) > 0$ for a correct classification.

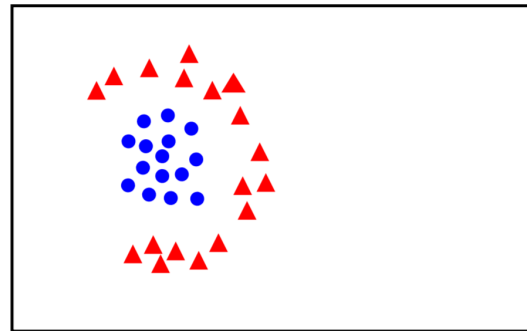
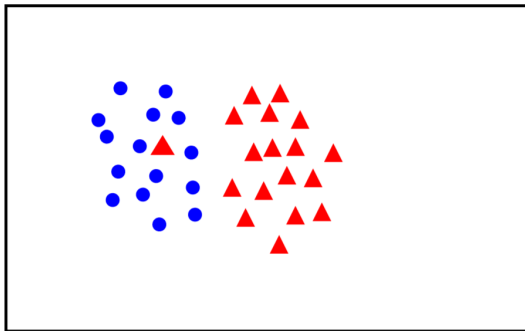


Linear separability

linearly
separable



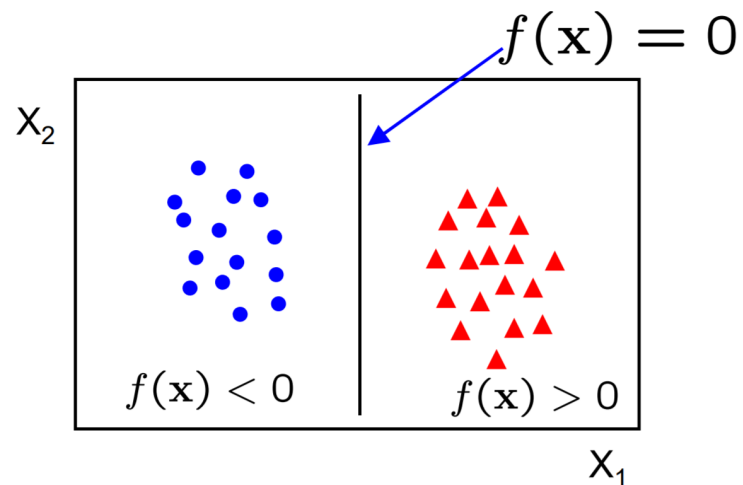
not
linearly
separable



Linear classifiers

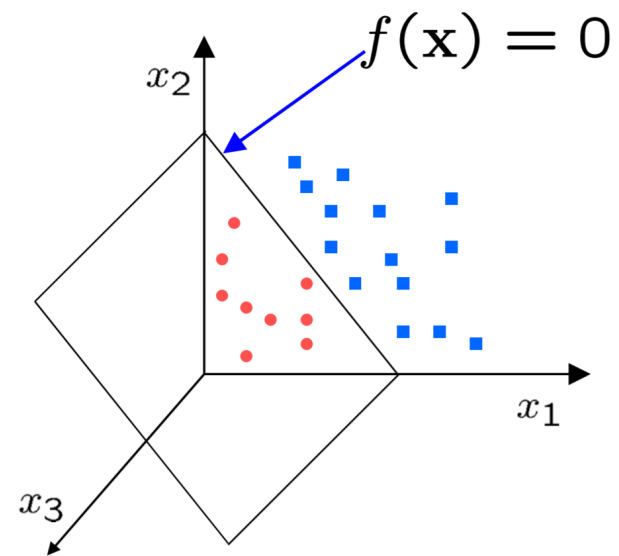
A linear classifier has the form

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b$$



- in 2D the discriminant is a line
- $\mathbf{w}$ is the **normal** to the line, and b the **bias**
- $\mathbf{w}$ is known as the **weight vector**

- in 3D the discriminant is a plane



Support Vector Machines

Note that

$$y = ax + b$$

is the same thing as

$$y - ax - b = 0$$

Given two vectors $\mathbf{w} = \begin{pmatrix} -b \\ -a \\ 1 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} 1 \\ x \\ y \end{pmatrix}$

$$\mathbf{w}^T \mathbf{x} = -b \times (1) + (-a) \times x + 1 \times y$$

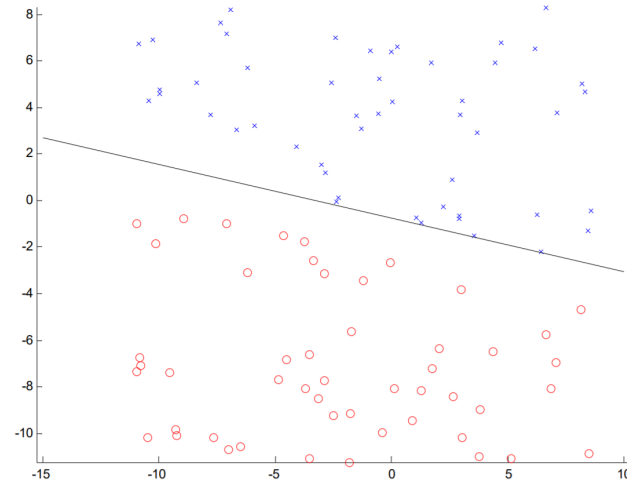
$$\mathbf{w}^T \mathbf{x} = y - ax - b$$

The two equations are just different ways of expressing the same thing.

<https://www.svm-tutorial.com/2014/11/svm-understanding-math-part-2/>

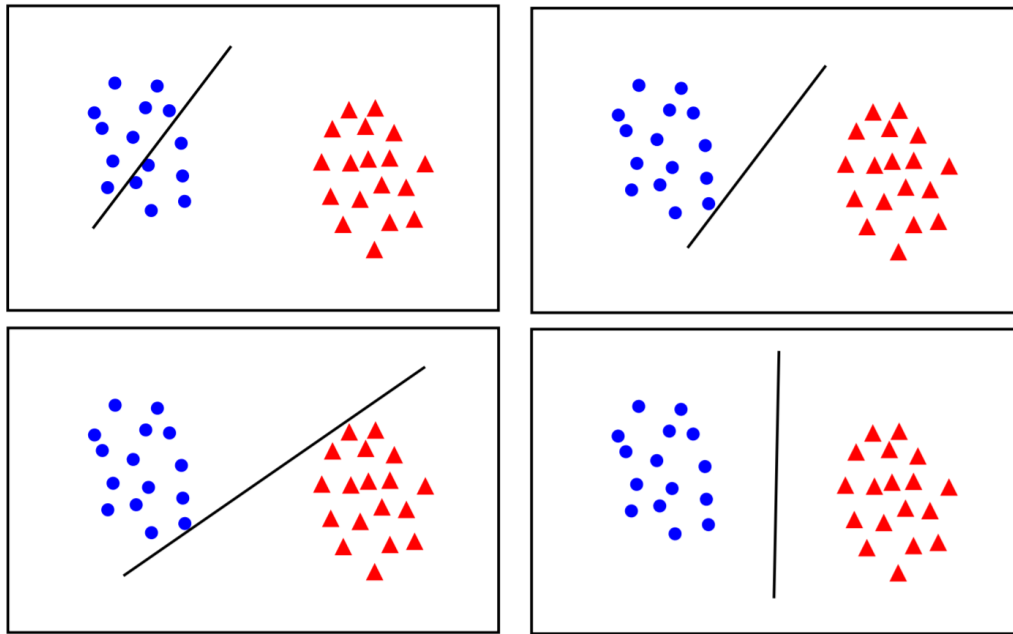
Support Vector Machines

example



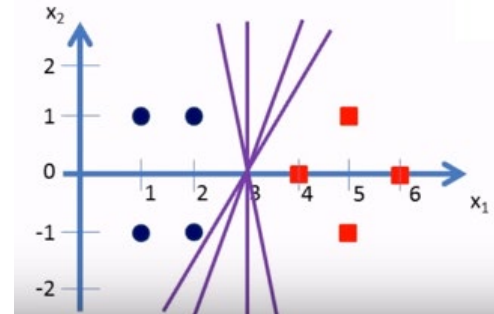
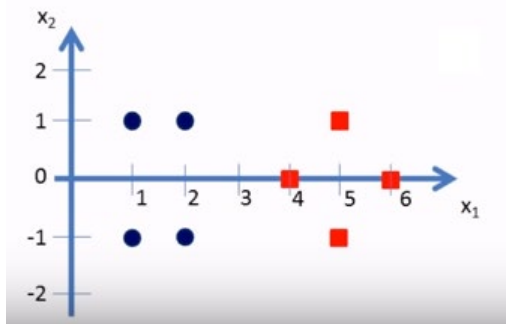
- if the data is linearly separable, then the algorithm will converge
- convergence can be slow ...
- separating line close to training data
- we would prefer a larger margin for generalization

What is the best w ?



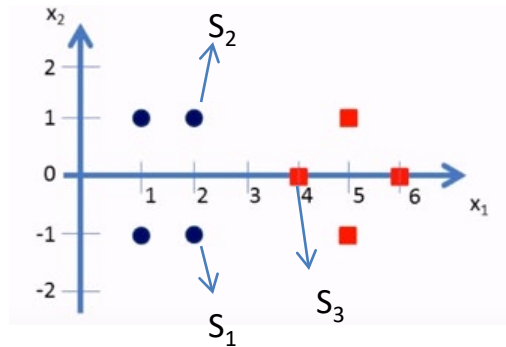
- **maximum margin** solution: most stable under perturbations of the inputs

Example:

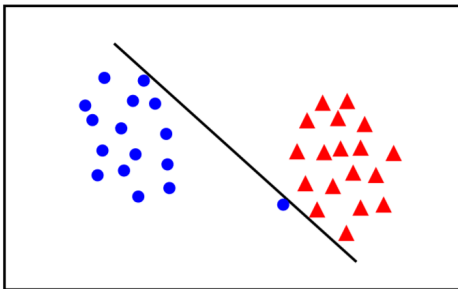


Example:

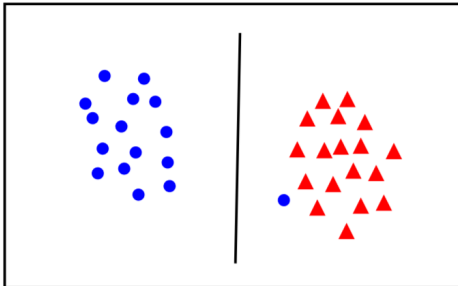
- Here, we select 3 Support Vectors to start with.
- They are S_1 , S_2 and S_3 .



Linear separability: What is the best w ?



- the points can be linearly separated but there is a very narrow margin

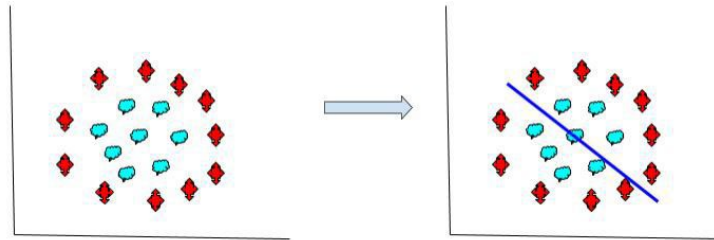


- but possibly the large margin solution is better, even though one constraint is violated

In general there is a trade off between the margin and the number of mistakes on the training data

Non-linearly Separable Data

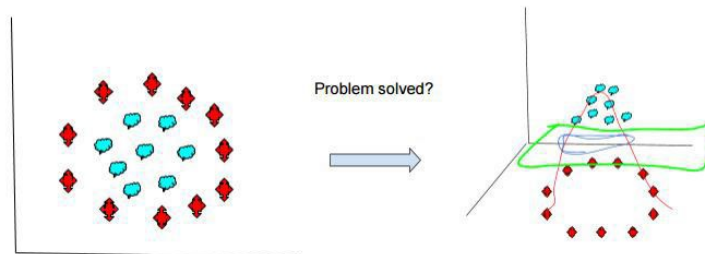
- SVM works well when the data points are linearly separable.
- If the decision boundary is non-linear then SVM may struggle to classify.
- SVM has no direct theory to set the non-linear decision boundary models.
- How does it handle the cases where the data is absolutely not linearly separable?



- This was not giving us good results. Now how do we get a non-linear decision boundary for the above data set? The answer is to use kernels.

Non-linearly Separable Data

- *Kernel:* This actually is a **function** that maps the data to a higher dimension where the data is separable. It transforms the linearly inseparable data into a linearly separable one by projecting it into a higher dimension.
- In simple words, the data is not linearly separable in this 2-dimension.
- Projecting the data up by some number of dimensions using a mapping function is useful. In that dimension plane the data will be separable.



Kernel Types

- Some popular choice for kernel functions are:
 - Gaussian Kernel
 - Polynomial Kernel

<https://www.mathworks.com/examples/statistics/mw/stats-ex18316588-train-svm-classifiers-using-a-gaussian-kernel>